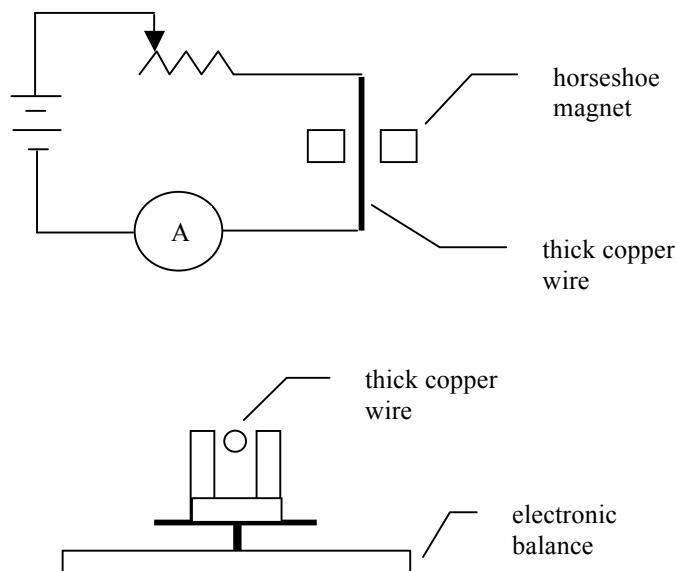


YEAR 12 PHYSICS
INVESTIGATION: STRENGTH OF A MAGNET

AIM To determine the magnetic field strength of a horseshoe magnet.

METHOD 1. Set up the following circuit.



2. Tare the balance to zero with the magnet in place.
3. Place the thick copper wire between the poles of the magnet.
4. Measure the width of the ends of the magnet and record it. This is taken as l .
5. Place a 1.00 A current through the wire and record the reading on the scale.
6. Repeat for increasing current values up to 5.00 A.
7. Graph the results of F against I and use this to determine the magnetic field strength of the magnet.

RESULTS

CURRENT (A)	MASS RECORDING ON SCALE (g)	FORCE EXERTED ON THE CONDUCTOR (N)

DISCUSSION Consider the following points.

- Approximations made.
- Sources of error.
- Accuracy of your result.

CONCLUSION Write a suitable conclusion.